Description for data used in the paper: "Stability of an Idealized Floating Carpet of Plastic Spheres in an Open Channel Flow"

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1 Introduction

This document describes the data obtained from laboratory experiments aimed to understand the stability of the debris accumulation, or the carpet, and flow dynamics under the carpet. The experiments were performed at Hydraulic Engineering Laboratory of Delft University of Technology, The Netherlands.

This dataset contains

1. Data for analysis of instability, squeezing and erosion, for the carpet (contained in `2cmCarpet.csv` file), and
2. Data for velocity measurements using Laser Doppler Anemometary (LDA) (contained in `2_00meterCarpet.pickle` file).

Experimental setup is shown in figure 1, showing three water level sensors, camera recording view and velocity measurement range of 3.0 meter. It is suggested to consult with Methods section of the published paper.

1.1 Data for analysis of instability of the carpet

This dataset was used for analysis of squeezing and erosion instabilities of the accumulation layer as shown in figure 2, or the carpet.

To load the data using `Python` programming language, run the following code

```
import pandas as pd
data = pd.read_csv('2cmCarpet.csv')
```

Loaded data includes

1. `Q (l/s)` : flow discharge measured in litre/second,
2. `depth (cm)` : water depth measured in centimetre,
3. `velocity (m/s)` : flow velocity measured in metre/second,

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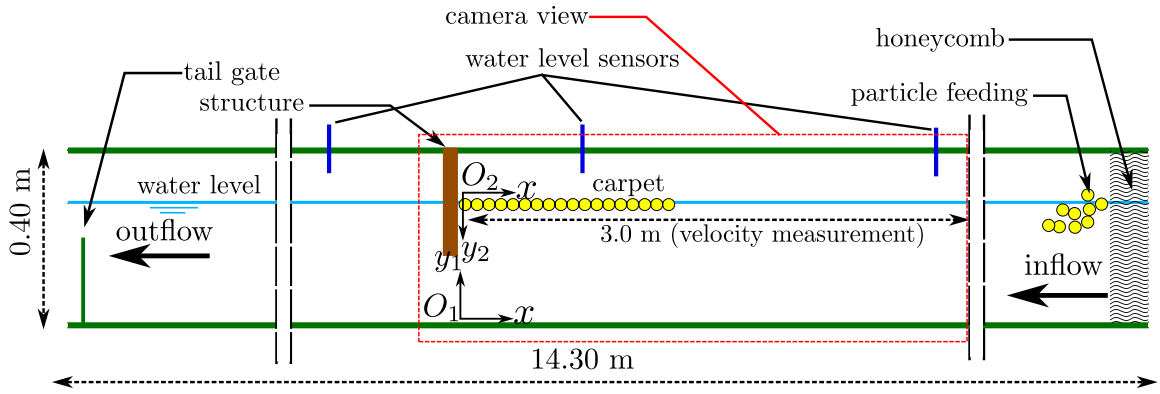
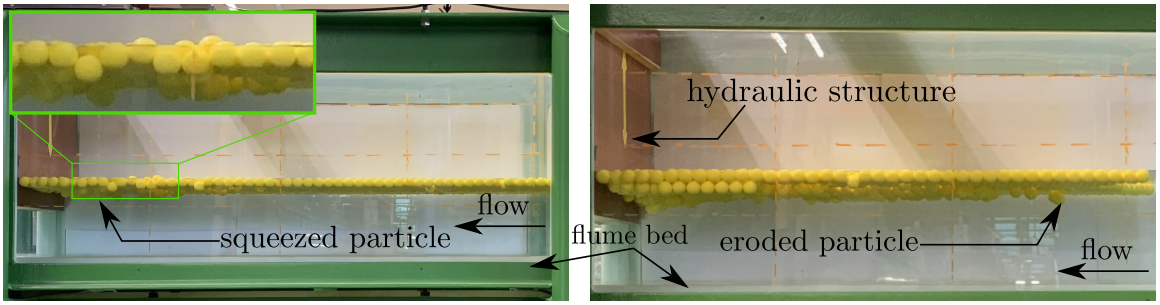


Figure 1: Experimental setup, showing three water level sensors, camera view and velocity measurement range of 3.0 m.



(a) Squeezing instability.

(b) Erosion instability.

Figure 2: Instability mechanisms of carpet: (a) squeezing, and (b) erosion.

4. `Length (m)` : carpet length measured in metre,
5. `squeeze` : squeezing situation of the carpet indicated by boolean vaules (0 or 1),
6. `erode` : erosion situation of the carpet indicated by boolean vaules (0 or 1)

1.2 Data for LDA velocity measurements

Flow velocity is measured using LDA system along the flume, starting at the position near the hydraulic structure shown in figure 1. Measurement was performed to obtain the vertical profile of flow velocity, and to gain insight of streamwise development of velocity profiles. Therefore, velocity was measured for every 25 cm along x_1-direction until 2.25 m (moving upstream direction), and one profile measurement at $x_1 = 3.0$ m.

To load the data file `2_00meterCarpet.pickle` in `Python`, please run the following command

```
with open('2_00meterCarpet.pickle', 'rb') as f: ### Loading the  data
        dataList = pickle.load(f)
```

If you want to have the data for position $x_1 = 0.25$ m upstream of the structure, please run

```
dataList[0]
```

where the index `0` can be changed depending on the measurement position.

Loaded data `dataList[0]` consists of relevant 9 columns:

1. `Date` : Date of doing experiment
2. `Time` : Time of doing experiment
3. `WaterLevel1 [V]` : Water level measured at $x_1 = 3.0$ m upstream of the structure, in voltage
4. `WaterLevel2 [V]` : Water level measured at $x_1 = 1.0$ m upstream of the structure, in voltage
5. `WaterLevel3 [V]` : Water level measured at $x_1 = 1.0$ m downstream of the structure, in voltage
6. `DisplacementLDA [V]` : Vertical position of the LDA system, measured in voltage
7. `FlowRate [volt]` : Flow discharge measured in voltage
8. `Vel1 [V]` : Signal A obtained from LDA, measured in voltage
9. `Vel2 [V]` : Signal B obtained from LDA, measured in voltage
10. `Unnamed:9` : Irrelevant column

1.2.1 Calibration process

1. To obtain water level [cm] at $x_1 = 3.0$ m upstream of the structure, please apply the following formula
$$38.01 - 3.76 \times \texttt{WaterLevel1 [V]} \tag{1}$$
2. To obtain water level [cm] at $x_1 = 1.0$ m upstream of the structure, please apply the following formula
$$38.18 - 3.71 \times \texttt{WaterLevel2 [V]} \tag{2}$$
3. To obtain water level [cm] at $x_1 = 1.0$ m downstream of the structure, please apply the following formula
$$36.42 - 3.8 \times \texttt{WaterLevel3 [V]} \tag{3}$$
4. To obtain the vertical position [cm] of LDA sensor measured from the bottom of the flume, please apply the following formula
$$25.753 - 2.4953 \times \texttt{DisplacementLDA [V]} \tag{4}$$
5. To obtain the flow discharge [l/s], please apply the following formula
$$-6.37 + 3.14 \times \texttt{FlowRate [volt] [V]} \tag{5}$$
6. To obtain the streamwise component of the velocity u_1 [m/s], please apply the following formula
$$0.096 \times (\texttt{Vel2 [V]} - \texttt{Vel1 [V]}) \tag{6}$$
7. To obtain the vertical component of the velocity u_2 [m/s], please apply the following formula
$$0.096 \times (\texttt{Vel2 [V]} + \texttt{Vel1 [V]}) \tag{7}$$